

with matching snippets, according to their needs.

Concurrent connections: Tornado offers real time services and supports a large number of concurrent connections, HTTP streaming (a communication protocol implemented by Apple Inc) and long polling (it is a 'push' technology, which allows an emulative push mechanism under circumstances in which a real push is not possible). With Tomado, it is very easy to write real time services. FriendFeed maintains an open connection, especially for its frequently involved users.

High performance: This is Tornado's most attractive feature. It is very fast compared to all other Python Web frameworks. Considering its baseline output, it is around four times higher in speed and pretty efficient also.

Getting started with Tornado

Tornado platforms: Tomado runs on most of the platforms like UNIX, Windows, and so on. For better performance and scalability, only Linux with EPoll (a Linux kernel system call which replaced older calls to achieve better performance), and BSD (Berkeley Standard Distribution) with KQueue (an event-notification interface) are recommended for effective production deployment. Windows is officially not supported but is recommended only for development purposes.

Installation: Tornado is listed in PyPI, which is a Python Package Index—a repository of software for the Python programming language. As Tomado is listed in PyPI, it can be installed easily using Pip:

```
pip install tornado
```

As an example, let's create a new project called webappdemo. We have the *app.py* python file whose contents are as follows:

app: It is the main method for Python to start executing and is the entry for the program.

tornado.ioloop: The *ioloop* class is the main event for any application which can create a non-blocking thread for an application.

tornado.web: This is the basic module of Tornado, which implements a lightweight Web development framework. It was built in FriendFeed using their non-blocking HTTP server and low-level I/O modules. In this demo, MainHandler is the single handler mapped to the root path.

MainHandler: This is a handler for a request, which refers to the *self.write()* method. The MainHandler class returns the response using this method.

application.listen(8888): We look at the specification 8888 first. It is just a port which the application considers (i.e., listens to). Any one can communicate with this application through this particular port (8888).

It's now time to run the application. You can run this simple application in a terminal (if you are using Ubuntu). If you are using the Windows platforms, you can run this application using the command prompt.

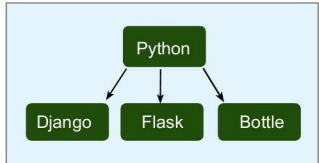


Figure 1: Python frameworks

To run your application in a terminal, use:

```
python app.py
```

Finally, you should be able to visit your website at <http://localhost:8888>

Tornado and other Python Web frameworks

Pyramid, Flask, Bottle and Django are among the well known frameworks of Python.

Pyramid is a small framework which makes real-world Web applications productive. Its advantages over other frameworks are:

- A quick start
- Flexibility
- Compatible with Python 3
- Ability to develop large Web applications, etc.

Flask is a micro framework of Python which is easy to set up and comes with a built-in server. Flask's main features are:

- It implements most commonly used core components
- Provides users a choice for database access or validation
- Flask extensions are available

Bottle is a framework that provides minimal tools out-of-the-box (routing, templating, etc).

It also comes under the category of 'micro', which enables compilation in Python 3. Some of the other features of Bottle are:

- Ability to build simple projects
- Efficient in creating Web application interfaces
- Flexibility

Django is a powerful Web framework that is supported by a large and active community. It offers model-based forms and has its own excellent documentation. Django makes it easier to build better Web applications, quickly and with less code. It is a high-level Python Web framework that encourages rapid development and perfect design.

Some of the attractive features of Django are:

- Predefined powerful admin interface
- Offers model-based forms
- Django extensions are available
- Has useful applications

The purpose of a Web framework is to provide some common features that are essential for all kinds of